

Finite Rational Reductions for the Convex-to-Polytope Conjecture in Neural Codes

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Abstract

A finite neural code is *open convex* if it records the atoms of a finite family of open convex receptive fields, and it is *polytope convex* if the receptive fields may be chosen as interiors of convex polytopes. Whether these two classes coincide in arbitrary finite dimension is open. We develop a finite reduction theory for this conjecture. First, we prove polytope monotonicity: adjoining arbitrary nonmaximal words beneath the same maximal codewords preserves polytope convexity. This yields an explicit polytopal realization of every max-intersection-complete code. We then show that universal polyhedralization is equivalent, as a universally quantified theorem scheme, to deleting one nonmaximal word from a polytopal code, and further reduces to deleting a binary meet $\sigma = \tau \cap \upsilon$ of strict surviving superwords. Every open-convex binary-meet deletion carries finite rational one-dimensional bridge data, including simultaneous endpoint events. We also prove that one arbitrary convex neuron over a fixed finite polytopal arrangement can always be replaced by one polytope in the same dimension. Finally, every bounded open convex realization admits a protected inner-polytopal supercode and a finite source-contained whole-word repair atlas for all unwanted atoms. Exact rational examples show why independent or sequential convexification of those repairs can fail. These results do not resolve the convex-to-polytope conjecture; they reduce it to a finite simultaneous attachment problem for repeated neurons.

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1 Introduction

A population code records which neurons are active together. If neuron i is active on a receptive field U_i inside a stimulus space X , then a codeword σ occurs when there is a point lying in exactly the receptive fields indexed by σ . In models of place-cell activity, the U_i are commonly idealized as convex regions. The resulting *convex neural codes* lie at an intersection of combinatorics, convex geometry, topology, and the geometry of neural representations; see Chen et al. [2], Cruz et al. [3], Curto et al. [4].

The present paper concerns a sharper geometric question. An open convex realization may have curved boundaries. Must the same finite atom code always be realizable by interiors of convex polytopes, perhaps after increasing the dimension? Kunin, Lienkaemper, and Rosen proved that

polytope-convex codes are exactly the minors of positive-covector codes of finite real-representable oriented matroids [8]. Thus a positive answer would place every open convex code inside a finite real hyperplane arrangement. The planar case is known: every planar open convex code admits a polygonal realization [1]. In higher dimension, the conjecture remains open [7, 8].

The difficulty is not the preservation of the nerve. A nerve records nonempty intersections, whereas a neural code records every relative atom. Inner approximation can remove memberships and create smaller codewords; outer approximation can add memberships and create larger ones. Boundary-supported and lower-dimensional atoms prevent one from assuming positive geometric margins. The main purpose of this paper is to isolate what can nevertheless be made finite and rational.

Main contributions

The results form one reduction chain.

- (i) **Polytope monotonicity.** If $\mathcal{A}$ is polytope convex and

$$\mathcal{A} \subseteq \mathcal{E} \subseteq \Delta(\mathcal{A}),$$

then $\mathcal{E}$ is polytope convex. The proof is an explicit finite cap construction with one independent height coordinate per added word. It includes threshold equality and simultaneous-cap analysis.

- (ii) **Polytopal maximal-intersection cores.** The code consisting of the empty word and all nonempty intersections of the maximal words has the affine-hyperplane realization of Cruz et al. [3], which we record in a bounded polytopal form. Combining this construction with polytope monotonicity yields polytope convexity for every max-intersection-complete code, and hence for every intersection-complete code. The contribution here is the bounded finite formulation and its role in the reduction chain, not priority for the underlying core construction.
- (iii) **Three universal reductions.** Universal polyhedralization is equivalent, as a theorem scheme, to lifting every immediate cover from a polytopal lower code. It is also equivalent to deleting one nonmaximal word from a polytopal code whenever the deletion is open convex. Finally, it suffices to treat deletions

$$\mathcal{E} = \mathcal{D} \setminus \{\sigma\}, \quad \sigma = \tau \cap \upsilon,$$

where τ and υ are strict surviving superwords. The last reduction uses an inclusion-minimal family of maximal words and a deletion order by nondecreasing cardinality.

- (iv) **Finite rational bridges.** For every open-convex binary-meet deletion, the segment between witnesses of τ and υ produces a covered interval code in the strict trunk above σ . Only the weak order and equality classes of finitely many interval endpoints matter, so the entire trace can be rationalized without changing simultaneous events.
- (v) **One-neuron polyhedralization over a fixed arrangement.** If the old receptive fields already form a finite polytopal arrangement, then one additional arbitrary open convex neuron can be replaced by one polytope in the same dimension while preserving every atom. The proof uses the finite face decomposition of the old arrangement and weak separation of every outside cell or outside witness.

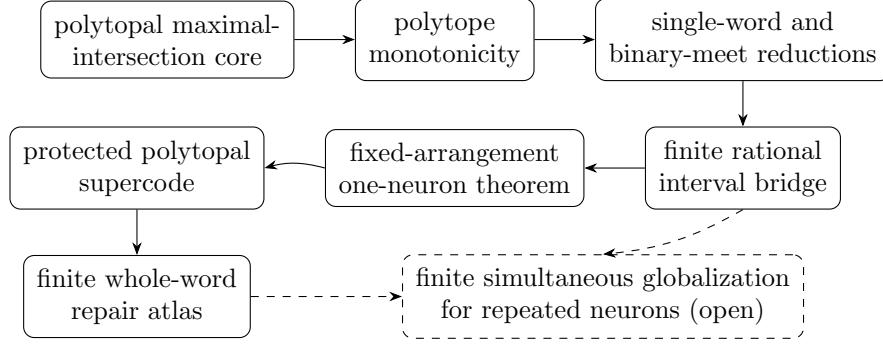


Figure 1: The theorem dependency chain. Solid arrows are proved implications. The dashed final arrow is the open simultaneous-globalization step.

- (vi) **Finite protected supercodes and repair atlases.** Every bounded open convex realization admits a finite inner-polytopal supercode $\mathcal{D}$ with

$$\mathcal{C} \subseteq \mathcal{D} \subseteq \Delta(\mathcal{C}),$$

the same maximal words, and an exact protected witness for every target word. Every unwanted carrier in $\mathcal{D} \setminus \mathcal{C}$ is covered by finitely many full-dimensional polytopes lying inside complete source words. Hence zero margin does not force infinitely many local repair charts.

The last statement is not a simultaneous-globalization theorem. We give an exact rational example in which repairing one word and then convexifying a second neuron recreates the first forbidden word; synchronized whole-word enlargement succeeds in the same example. This proves that the remaining step is a genuine finite compatibility problem rather than a routine convex-hull operation.

Relation to prior work

Open-convex monotonicity and open/closed convexity of max-intersection-complete codes were established by Cruz et al. [3]. Morphisms and the code-minor poset were developed by Jeffs [5], with an erratum in Jeffs [6]; the covering relations were later classified by Jeffs and Trang [7]. The representable-oriented-matroid characterization of polytope-convex codes is due to Kunin et al. [8], and planar polygonalization to Bukh and Jeffs [1]. Our contributions are the polytopal cap upgrade and its combination with the known core realization, the precise deletion and binary-meet reduction chain, rational bridge extraction, fixed-arrangement one-neuron theorem, and the protected-supercode/finite-atlas package. The priority status of each theorem is recorded in the accompanying audit.

Organization and scope

Section 2 fixes conventions. Section 3 gives the cap construction and maximal-intersection cores. Section 4 proves the three reductions. Section 5 constructs and rationalizes interval bridges. Section 6 proves one-neuron polyhedralization. Sections 7 and 8 produce the finite protected supercode and repair atlas. Section 9 gives exact failure calibrations and states the remaining theorem. The appendices contain the convention crosswalk, full cap audit, a known intersection-morphism fact that rules out a tempting shortcut, and verifier specifications.

The central thesis is therefore precise: the convex-to-polytope conjecture reduces to a canonical binary-meet deletion operation with finite rational bridge data; local curvature of one neuron over a fixed arrangement is never the obstruction; and arbitrary realizations admit finite protected supercodes with finite repair atlases. What remains is global simultaneous convex attachment of those finite data while every repeated original neuron remains one convex polytope.

2 Conventions and preliminaries

Fix $[n] = \{1, \dots, n\}$. A *neural code* is a finite set

$$\mathcal{C} \subseteq 2^{[n]}$$

containing $\emptyset$. The mandatory empty word is convenient for the code-minor category and for ambient regions outside all receptive fields.

Definition 2.1 (Atoms and codes). Let X be an open convex subset of a finite-dimensional affine space and let $U_1, \dots, U_n \subseteq X$ be open convex sets. For $\sigma \subseteq [n]$, define

$$A_\sigma(\mathcal{U}, X) = X \cap \bigcap_{i \in \sigma} U_i \setminus \bigcup_{j \notin \sigma} U_j.$$

The code of the realization is

$$\text{code}(\mathcal{U}, X) = \{\sigma \subseteq [n] : A_\sigma(\mathcal{U}, X) \neq \emptyset\}.$$

The code is *open convex* if it has such a realization. It is *open polytope convex*, or simply *polytope convex*, if

$$U_i = \text{int } P_i, \quad X = \text{int } Q$$

for finite-dimensional convex polytopes P_i, Q , with interiors taken in the common ambient affine hull.

A neuron may be trivial, duplicate, full, or redundant. Trivial inactive neurons may be represented by lower-dimensional polytopes with empty ambient interior; a full neuron may be represented by the ambient polytope itself. We retain such coordinates when they matter combinatorially, but remove and restore them during constructions when this simplifies dimension statements.

Definition 2.2 (Simplicial completion and maximal words). The simplicial completion of $\mathcal{C}$ is

$$\Delta(\mathcal{C}) = \{\sigma : \sigma \subseteq \tau \text{ for some } \tau \in \mathcal{C}\}.$$

The inclusion-maximal words of $\mathcal{C}$ are denoted $\text{Max}(\mathcal{C})$. The code is *max-intersection-complete* if every nonempty intersection of maximal words belongs to $\mathcal{C}$; by convention $\emptyset \in \mathcal{C}$ as well.

Definition 2.3 (Trunks and morphisms). For $\sigma \subseteq [n]$, the trunk rooted at σ is

$$\text{Tk}_{\mathcal{C}}(\sigma) = \{\tau \in \mathcal{C} : \sigma \subseteq \tau\}.$$

A proper trunk is nonempty and not all of $\mathcal{C}$. A code morphism is a map whose inverse image of every proper trunk is a proper trunk. We write $\mathcal{C} \leq \mathcal{D}$ when there is a surjective morphism $\mathcal{D} \twoheadrightarrow \mathcal{C}$.

The minor convention used here is the mandatory-empty-word version of the poset introduced by Jeffs [5]. The exact operation-by-operation comparison with the older convention is given in Section A. We use two established facts: surjective code morphisms preserve open convexity [5], and polytope-convex codes are exactly the minors of positive-covector codes of finite real-representable oriented matroids [8].

Lemma 2.4 (Bounded realization). *Every finite open convex neural code has a bounded open convex realization in the same dimension.*

Proof. Choose one witness $x_\sigma \in A_\sigma$ for every $\sigma \in \mathcal{C}$, including $x_\emptyset$, and put

$$K = \text{conv}\{x_\sigma : \sigma \in \mathcal{C}\}.$$

The compact set K lies in the open convex ambient set X . Hence, for some $\varepsilon > 0$,

$$K + \varepsilon \overline{B} \subseteq X.$$

Set

$$X' = \text{int}(K + \varepsilon \overline{B}), \quad U'_i = U_i \cap X'.$$

Restriction cannot create a word, and every selected witness lies in X' . Therefore the code is unchanged; the empty atom is retained; and the new ambient is bounded, open, and convex. $\square$

Remark 2.5 (Strict interiors). The code depends on strict membership. A point on a facet of P_i is absent from the neuron $\text{int } P_i$. This convention is used explicitly in the cap-boundary and interval-endpoint arguments below.

3 Polytope monotonicity and polytopal cores

The open-convex monotonicity theorem of Cruz et al. [3] states that adding nonmaximal words beneath fixed maximal words preserves open convexity. Their work also gives the standard hyperplane-arrangement realization of the maximal-intersection core. We retain that core construction in bounded polytopal form and prove the finite cap counterpart of monotonicity needed throughout this paper.

Theorem 3.1 (Polytope monotonicity). *Let $\mathcal{A}$ be polytope convex and suppose*

$$\mathcal{A} \subseteq \mathcal{E} \subseteq \Delta(\mathcal{A}).$$

Then $\mathcal{E}$ is polytope convex. If $r = |\mathcal{E} \setminus \mathcal{A}|$, the construction uses r additional coordinates and one polyhedral cap for each added word.

Proof. Remove trivial inactive coordinates and restore them at the end. Start with

$$\mathcal{A} = \text{code}(\{\text{int } P_i\}_{i=1}^n, \text{int } K)$$

in $\mathbb{R}^d$, where K is a full-dimensional polytope containing one chosen witness from every atom. Enumerate

$$\mathcal{E} \setminus \mathcal{A} = \{\sigma_1, \dots, \sigma_r\}.$$

For each s , choose a maximal word $m_s \in \mathcal{A}$ containing σ_s . The atom A_{m_s} is open in the ambient: if a point in all fields indexed by m_s belonged to an additional field, its word would strictly contain m_s . Choose

$$a_s \in A_{m_s}, \quad \rho_s > 0, \quad \overline{B}(a_s, \rho_s) \subset A_{m_s} \cap \text{int } K.$$

Work in $\mathbb{R}^{d+r}$ with coordinates $(x, z_1, \dots, z_r)$. Put

$$B_0 = K \times [-1, 1]^r, \quad v_s = (a_s, 4e_s), \quad Q' = \text{conv}(B_0 \cup \{v_1, \dots, v_r\}).$$

Let $\pi(x, z) = x$ and define preliminary old-neuron polytopes

$$\tilde{P}_i = Q' \cap \pi^{-1}(P_i).$$

Every interior point of Q' projects to $\text{int } K$, and membership in $\text{int } \tilde{P}_i$ depends only on the old projection. Hence the preliminary code remains $\mathcal{A}$, and every old witness survives at height zero.

Choose $1 < h_s < 4$ and set

$$\lambda_s = \frac{h_s - 1}{3}$$

so that

$$\lambda_s > \frac{1}{2}, \quad (1 - \lambda_s) \text{diam}(K) < \rho_s.$$

If

$$y = b_0 p_0 + \sum_{t=1}^r b_t v_t, \quad b_t \geq 0, \quad b_0 + \sum_t b_t = 1, \quad p_0 \in B_0,$$

then the s th height satisfies

$$z_s(y) \leq b_0 + 4b_s \leq 1 + 3b_s.$$

Consequently

$$z_s(y) \geq h_s \implies b_s \geq \lambda_s.$$

All projected vertices lie in K , so

$$\|\pi(y) - a_s\| \leq (1 - b_s) \text{diam}(K) \leq (1 - \lambda_s) \text{diam}(K) < \rho_s.$$

Thus the entire closed cap $Q' \cap \{z_s \geq h_s\}$ projects into the maximal atom A_{m_s} . Two closed caps are disjoint, because simultaneous activation of caps $s \neq t$ would imply $b_s, b_t > 1/2$.

Define the final polytopes by

$$P'_i = \tilde{P}_i \cap \bigcap_{\{s: i \notin \sigma_s\}} \{z_s \leq h_s\}.$$

Below all thresholds the old word is unchanged. If $z_s > h_s$, the preliminary word is m_s , and the s th family of cuts removes exactly the neurons in $m_s \setminus \sigma_s$. Every other cap threshold is strict, so the final word is σ_s .

At equality $z_s = h_s$, the projection still lies in A_{m_s} . A neuron with $i \notin \sigma_s$ lies on a supporting hyperplane of P'_i and is therefore absent from its interior; a neuron in σ_s is uncut at this threshold and remains interior because the projection lies in the open maximal atom. No second threshold can be active. Hence the boundary word is again σ_s , not a new intermediate word.

Every cap contains an interior point of Q' above its threshold, so every new word occurs. Every old witness remains at height zero. The code is exactly $\mathcal{E}$. $\square$

Remark 3.2 (Why the coefficient $1/2$ matters). The cap argument is not a heuristic “one apex dominates” statement. Equation $b_s > 1/2$ is used only after the height inequality forces it. It implies pairwise cap disjointness even when a point has nonzero coefficients at many apices and no coefficient is initially distinguished. The equality case is treated by the strict-interior convention rather than discarded as measure zero.

Corollary 3.3 (Open-convex monotonicity). *If $\mathcal{A}$ is open convex and*

$$\mathcal{A} \subseteq \mathcal{E} \subseteq \Delta(\mathcal{A}),$$

then $\mathcal{E}$ is open convex.

Proof. First use Theorem 2.4 to restrict to a bounded realization inside a polytope K containing all witnesses. Repeat the cap construction, replacing the preliminary polytopes by inverse images of the open convex traces and replacing each closed cut by the corresponding open halfspace. The projection and disjoint-cap analysis is unchanged. This recovers the known monotonicity theorem of Cruz et al. [3] in the present convention. $\square$

3.1 The maximal-intersection core

Let $m_1, \dots, m_k$ be the maximal nonempty words of $\mathcal{C}$. Define

$$\mathcal{B}(\mathcal{C}) = \{\emptyset\} \cup \left\{ \bigcap_{a \in R} m_a : \emptyset \neq R \subseteq [k] \right\},$$

where duplicate intersections are retained only once.

Proposition 3.4 (Explicit polytopal core). *The code $\mathcal{B}(\mathcal{C})$ has a finite polytopal realization. If $k \geq 2$, it may be realized in dimension $k - 1$.*

Proof. Work in the affine hyperplane

$$H = \left\{ x \in \mathbb{R}^k : \sum_{a=1}^k x_a = 1 \right\}.$$

For each nontrivial neuron i , let

$$R_i = \{a \in [k] : i \in m_a\}$$

and define the open polyhedral field

$$U_i = \{x \in H : x_b < 0 \text{ for every } b \notin R_i\}.$$

For $x \in H$, put

$$\rho(x) = \{a : x_a \geq 0\}.$$

The set $\rho(x)$ is nonempty because the coordinates sum to 1. Moreover,

$$i \in c(x) \iff \rho(x) \subseteq R_i \iff i \in \bigcap_{a \in \rho(x)} m_a.$$

Every nonempty subset $R \subseteq [k]$ occurs as $\rho(x)$: choose positive coordinates on R and sufficiently small negative coordinates off R , then normalize the sum to 1.

Choose one witness for each distinct nonempty intersection word and a full-dimensional polytope $K \subset H$ whose interior contains them. Let $Q \supsetneq K$ be a larger full-dimensional polytope and put

$$P_i = K \cap \bigcap_{b \notin R_i} \{x_b \leq 0\}.$$

Inside $\text{int } K$, the strict inequalities give exactly the nonempty intersections of maximal words. Every point of $\text{int } Q \setminus K$ belongs to no $\text{int } P_i$ and supplies the empty word. This realizes $\mathcal{B}(\mathcal{C})$. The case $k = 1$ uses one common inner interval for all neurons of m_1 inside a larger ambient interval. $\square$

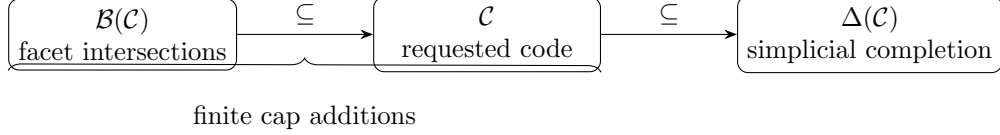


Figure 2: The maximal-intersection core and polytope monotonicity. The core is realized explicitly; every intermediate code with the same maximal words is obtained by finite caps.

Theorem 3.5 (Max-intersection-complete codes are polytopal). *Every finite max-intersection-complete neural code is polytope convex. In particular, every intersection-complete code is polytope convex.*

Proof. Let $\mathcal{C}$ be max-intersection-complete. Its core $\mathcal{B}(\mathcal{C})$ has the same maximal words, and

$$\mathcal{B}(\mathcal{C}) \subseteq \mathcal{C} \subseteq \Delta(\mathcal{B}(\mathcal{C})).$$

Apply Theorem 3.1. An intersection-complete code is max-intersection-complete. □

Remark 3.6 (Priority and role). The open and closed convexity of max-intersection-complete codes, together with an embedding-dimension bound, is due to Cruz et al. [3]. The construction above extracts an explicit bounded polytopal core and combines it with the cap theorem. We use the result as the positive starting point for the deletion reductions, not as a characterization of all convex codes.

4 Universal reductions to binary-meet deletion

This section distinguishes three logically equivalent *universal theorem schemes*. The equivalence does not mean that every individual nonmaximal deletion is itself a cover or a binary meet.

4.1 Immediate covers

Lemma 4.1 (Finite lower intervals). *For every finite code $\mathcal{C}$, the interval*

$$[\{\emptyset\}, \mathcal{C}] = \{\mathcal{D} : \mathcal{D} \leq \mathcal{C}\} / \cong$$

is finite.

Proof. A nontrivial output coordinate of a surjective morphism from $\mathcal{C}$ is determined by the inverse image of its simple trunk, which is a proper trunk of $\mathcal{C}$. After deleting trivial coordinates and identifying duplicate coordinates, an image is determined, up to permutation, by a finite set of distinct proper trunks. If $t(\mathcal{C})$ is the number of proper trunks, there are at most $2^{t(\mathcal{C})}$ such sets. Further removal of redundant coordinates can only decrease the count. □

Lemma 4.2 (Morphic preservation). *A surjective code morphic image of an open convex code is open convex. A surjective morphic image of a polytope-convex code is polytope convex.*

Proof. For each nontrivial output coordinate j , write the inverse image of its simple trunk as $\text{Tk}_{\mathcal{C}}(\rho_j)$ and define

$$V_j = \bigcap_{i \in \rho_j} U_i.$$

Then $x \in V_j$ exactly when the old word at x belongs to the defining trunk, so the new word is the morphic image of the old word. Intersections of open convex sets are open convex. If the old fields are interiors of polytopes, replace V_j by the interior of $\bigcap_{i \in \rho_j} P_i$; nonemptiness of the trunk ensures the usual interior-intersection identity. The mandatory empty word belongs to no proper trunk. $\square$

Theorem 4.3 (Saturated-chain reduction). *Universal polyhedralization is equivalent to the following local theorem scheme:*

$$\mathcal{A} \triangleleft \mathcal{B}, \quad \mathcal{A} \text{ polytope convex}, \quad \mathcal{B} \text{ open convex} \implies \mathcal{B} \text{ polytope convex.} \quad (\text{CL})$$

Proof. Universal polyhedralization plainly implies (CL). Conversely, the finite interval below an open convex code $\mathcal{C}$ contains a saturated chain

$$\{\emptyset\} = \mathcal{C}_0 \triangleleft \mathcal{C}_1 \triangleleft \cdots \triangleleft \mathcal{C}_m = \mathcal{C}.$$

Every $\mathcal{C}_s$ is a minor of $\mathcal{C}$ and hence open convex by Theorem 4.2. The one-word code is polytopal. Repeated application of (CL) proves every $\mathcal{C}_s$ polytopal. $\square$

The covering relations in the code poset have now been classified by Jeffs and Trang [7]. Our later reductions do not require the four-type classification; they pass instead through monotonicity and maximal-word intersections.

4.2 Single nonmaximal deletion

Let $\widehat{\mathcal{C}}$ denote the intersection completion generated by the maximal words of $\mathcal{C}$ together with all requested words; equivalently, it is the smallest intersection-complete code containing $\mathcal{C}$ without changing the maximal words.

Theorem 4.4 (Single-codeword-deletion reduction). *Universal polyhedralization is equivalent to the theorem scheme:*

$$\text{If } \mathcal{D} \text{ is polytope convex, } \mathcal{E} = \mathcal{D} \setminus \{\sigma\} \text{ is open convex, and } \sigma \text{ is nonmaximal in } \mathcal{D}, \text{ then } \mathcal{E} \text{ is polytope convex.} \quad (\text{DEL})$$

Proof. Universal polyhedralization implies (DEL). Conversely, fix an open convex $\mathcal{C}$. By Theorem 3.5, its intersection completion $\widehat{\mathcal{C}}$ is polytope convex. Delete the finitely many words of $\widehat{\mathcal{C}} \setminus \mathcal{C}$ one at a time. Every intermediate code lies between $\mathcal{C}$ and $\widehat{\mathcal{C}}$ and has the same maximal words, so it is open convex by Theorem 3.3. Every deleted word is nonmaximal because all maximal words already lie in $\mathcal{C}$. Applying (DEL) at each step ends at a polytopal realization of $\mathcal{C}$. $\square$

4.3 Binary meets

Define

$$\mathcal{B}(\mathcal{C}) = \{\emptyset\} \cup \left\{ \bigcap_{a \in R} m_a : \emptyset \neq R \subseteq [k] \right\},$$

where $m_1, \dots, m_k$ are the maximal words of $\mathcal{C}$, and put

$$\mathcal{C}^+ = \mathcal{C} \cup \mathcal{B}(\mathcal{C}).$$

Theorem 4.5 (Binary-meet deletion reduction). *Let $\mathcal{C}$ be open convex. Then $\mathcal{C}^+$ is polytope convex. Moreover, the words of $\mathcal{B}(\mathcal{C}) \setminus \mathcal{C}$ may be ordered as $\sigma_1, \dots, \sigma_r$ so that, with*

$$\mathcal{D}_0 = \mathcal{C}^+, \quad \mathcal{D}_s = \mathcal{D}_{s-1} \setminus \{\sigma_s\},$$

every $\mathcal{D}_s$ is open convex and, at the s th deletion, there are strict surviving superwords $\tau_s, v_s \in \mathcal{D}_s$ with

$$\sigma_s = \tau_s \cap v_s.$$

Consequently, universal polyhedralization is equivalent to proving all open-convex binary-meet deletions from polytopal lower codes are polytopal.

Proof. The core $\mathcal{B}(\mathcal{C})$ is polytope convex, and

$$\mathcal{B}(\mathcal{C}) \subseteq \mathcal{C}^+ \subseteq \Delta(\mathcal{B}(\mathcal{C})).$$

Thus $\mathcal{C}^+$ is polytope convex by monotonicity. It is also open convex because

$$\mathcal{C} \subseteq \mathcal{C}^+ \subseteq \Delta(\mathcal{C}).$$

Order the missing core words by nondecreasing cardinality, breaking ties arbitrarily. Fix the current word σ . Choose an inclusion-minimal nonempty family $R \subseteq \text{Max}(\mathcal{C})$ satisfying

$$\sigma = \bigcap_{m \in R} m.$$

The family has at least two members, because a one-member intersection would be a maximal word already in $\mathcal{C}$. Inclusion minimality implies

$$\bigcap_{m \in R'} m \supsetneq \sigma$$

for every proper nonempty $R' \subsetneq R$.

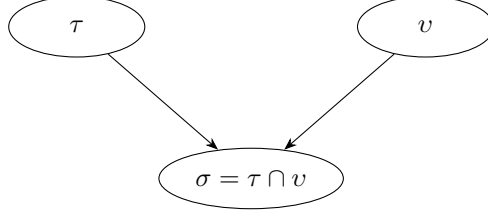
Partition $R = A \sqcup B$ into two nonempty subfamilies and put

$$\tau = \bigcap_{m \in A} m, \quad v = \bigcap_{m \in B} m.$$

Then $\tau, v \supsetneq \sigma$ and $\tau \cap v = \sigma$. Their cardinalities are strictly larger than $|\sigma|$. If one is absent from the original code, it appears later in the nondecreasing-cardinality order and has not yet been deleted; if it lies in $\mathcal{C}$, it is never deleted. Hence both survive the current deletion.

Every intermediate code lies between $\mathcal{C}$ and $\mathcal{C}^+$ with unchanged maximal words, so open-convex monotonicity applies. A universal binary-meet deletion theorem therefore yields polytopality at every step. The converse is immediate from universal polyhedralization. $\square$

Remark 4.6 (Logical strength). The theorem is a reduction of universal statements. It does not assert that an arbitrary individual nonmaximal word is a binary meet in the code from which it is deleted. The binary-meet property is guaranteed only along the canonical cardinality-ordered deletion path from $\mathcal{C}^+$ to $\mathcal{C}$.



delete σ ; keep the two strict superwords

Figure 3: A binary-meet deletion. The absent word σ is the intersection of two strict surviving superwords τ and v .

5 Finite rational bridges for binary-meet deletions

The binary-meet reduction is combinatorial. Open convexity supplies additional one-dimensional geometry that is both finite and exactly rationalizable.

Definition 5.1 (Strict relative trunk). For any $\sigma \subseteq [n]$, define

$$\text{ST}_{\mathcal{C}}(\sigma) = \{\emptyset\} \cup \{\rho \setminus \sigma : \rho \in \mathcal{C}, \sigma \subsetneq \rho\}.$$

Theorem 5.2 (Rational interval bridge). *Let*

$$\mathcal{E} = \mathcal{D} \setminus \{\sigma\}$$

be open convex and suppose $\tau, v \in \mathcal{E}$ are strict superwords with $\sigma = \tau \cap v$. Then there are a bounded open interval T , open intervals $J_j \subseteq T$ for $j \notin \sigma$ (possibly empty or all of T), and marked points $t_\tau, t_v \in T$ such that:

(i) *the J_j cover T ;*

(ii) *every relative word*

$$\rho(t) = \{j \notin \sigma : t \in J_j\}$$

satisfies $\sigma \cup \rho(t) \in \mathcal{E}$;

(iii) *the marked words are $\tau \setminus \sigma$ and $v \setminus \sigma$;*

(iv) *the relative empty word never occurs on T ;*

(v) *T , every endpoint, and both marked points may be chosen rational without changing the complete trace code, including simultaneous events.*

Thus the mandatory-empty-word interval code

$$\{\emptyset\} \cup \{\rho(t) : t \in T\}$$

is a rational subcode of $\text{ST}_{\mathcal{E}}(\sigma)$.

Proof. Choose witnesses

$$x \in A_\tau, \quad y \in A_v$$

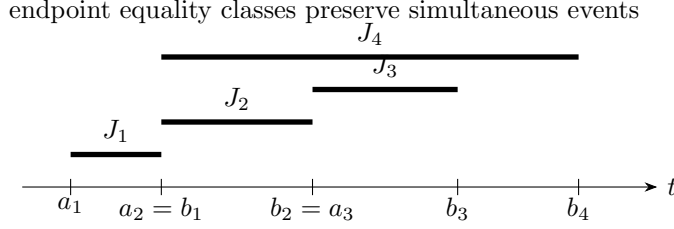


Figure 4: A rational interval bridge. Vertical marks are endpoint equality classes; several neurons may change simultaneously. Coverage prevents the relative empty word.

in an open convex realization $(X; U_1, \dots, U_n)$ of $\mathcal{E}$. Let

$$z(t) = (1 - t)x + ty.$$

Because $x, y \in X$ and $x, y \in U_i$ for every $i \in \sigma$, convexity gives

$$z([0, 1]) \subseteq X \cap \bigcap_{i \in \sigma} U_i.$$

The σ atom is absent. Hence every point of this compact segment belongs to some U_j with $j \notin \sigma$. The inverse image under z of

$$\left(X \cap \bigcap_{i \in \sigma} U_i \right) \cap \bigcup_{j \notin \sigma} U_j$$

is an open subset of $\mathbb{R}$ containing $[0, 1]$. Choose a bounded open interval T with $[0, 1] \subset T$ inside that inverse image.

For $j \notin \sigma$, set

$$J_j = T \cap z^{-1}(U_j).$$

The inverse image of an open convex set under an affine map is an open convex subset of a line, hence an open interval, possibly empty or all of T . The intervals cover T . At any $t \in T$, the full word of $z(t)$ is a word of $\mathcal{E}$, contains σ , and strictly contains it because at least one J_j covers t . At $t = 0$ and $t = 1$, the words are τ and v .

For rationalization, record the two endpoints of T , the endpoints of every nonempty proper J_j , and the two marked points. These form a finite total preorder. Equal endpoints, simultaneous starts or ends, and marked points in the same equality class are retained together. Map the distinct classes order-preservingly to rational numbers. Membership in an open interval depends only on the strict order of a test point relative to its endpoint classes. Thus every open cell and every endpoint class has the same membership vector before and after the reparameterization. $\square$

Corollary 5.3 (Finite event path). *The bridge words form a finite path of endpoint events. Between consecutive endpoint classes the word is constant; at one endpoint class all intervals beginning or ending there change simultaneously. No genericity or one-event-at-a-time assumption is required.*

5.1 An explicit finite bridge module

The theorem above is trace data rather than a global attachment theorem. It can nevertheless be realized as an independent finite polytopal module. After an affine rescaling take $T = (0, 1)$. For every proper interval $J_j = (a_j, b_j)$ choose the concave tent

$$h_j(t) = \min\{t - a_j, b_j - t\};$$

use a positive constant for $J_j = T$ and a negative constant for $J_j = \emptyset$. Then $J_j = \{h_j > 0\}$. The hypograph description of the tents gives a finite rational polyhedral realization in a two-dimensional strip, and a small simplex factor makes every protected trace word full dimensional if desired.

Remark 5.4 (The remaining attachment problem). The bridge is finite, one-dimensional, rational, and covered. These properties do not imply that it can be inserted into an arbitrary polytopal lower realization while every old occurrence of each repeated neuron remains part of one convex polytope. That global insertion problem is precisely what remains open after the reductions in this paper.

6 One-neuron polyhedralization over a fixed arrangement

The next theorem eliminates local curvature once the old realization has already been made polytopal.

Lemma 6.1 (Open-convex separation). *Let $U \subseteq \mathbb{R}^d$ be nonempty, open, and convex, and let $B \subseteq \mathbb{R}^d$ be nonempty and convex with $U \cap B = \emptyset$. Then there is an affine functional ℓ such that*

$$\ell(u) < 0 \quad (u \in U), \quad \ell(b) \geq 0 \quad (b \in B).$$

Proof. Because U is open, $U \cap \overline{B} = \emptyset$: otherwise an open neighborhood inside U would meet B . Apply the separating-hyperplane theorem to U and $\overline{B}$, choosing the open side to contain U . $\square$

Theorem 6.2 (Fixed-arrangement one-neuron polyhedralization). *Suppose*

$$\mathcal{D} = \text{code}(\{\text{int } P_i\}_{i=1}^n, \text{int } Q)$$

for convex polytopes $P_1, \dots, P_n, Q \subset \mathbb{R}^d$. Let $U \subseteq \text{int } Q$ be nonempty, open, and convex, and let $\mathcal{C}$ be the code obtained by adjoining U as a new neuron α . Then there is a convex polytope $P_\alpha \subseteq Q$ such that replacing U by $\text{int } P_\alpha$ leaves the complete atom code unchanged. The dimension does not increase.

Proof. Replace every old polytope by $P_i \cap Q$. Delete code-trivial coordinates whose intersections have empty ambient interior, and restore them afterward by lower-dimensional polytopes. The remaining old polytopes and Q have finitely many facet hyperplanes.

For an old word $\tau \in \mathcal{D}$, let A_τ denote its old atom. Partition the old words into

$$\mathcal{D}_+ = \{\tau : A_\tau \subseteq U\},$$

$$\mathcal{D}_- = \{\tau : A_\tau \cap U = \emptyset\},$$

and

$$\mathcal{D}_\pm = \{\tau : A_\tau \cap U \neq \emptyset, A_\tau \setminus U \neq \emptyset\}.$$

These cases say, respectively, that only $\tau\alpha$, only τ , or both words occur in the extension.

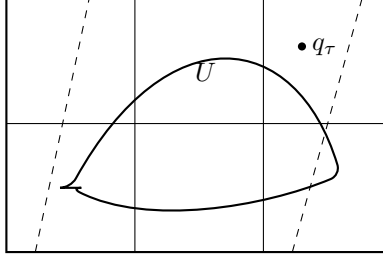
Take the finite affine arrangement formed by all facet hyperplanes of $Q, P_1, \dots, P_n$. The relative interiors of its faces inside $\text{int } Q$ form a finite partition. Membership in every old field is constant on each face, so every old atom is a finite union of relatively open convex polyhedral cells.

For every $\tau \in \mathcal{D}_-$, include all face cells of A_τ in a finite family $\mathcal{F}$. For every $\tau \in \mathcal{D}_\pm$, choose

$$p_\tau \in A_\tau \cap U, \quad q_\tau \in A_\tau \setminus U,$$

and include the singleton $\{q_\tau\}$ in $\mathcal{F}$. Every $B \in \mathcal{F}$ is convex and disjoint from U . By Theorem 6.1, choose an affine functional ℓ_B with

$$U \subseteq \{\ell_B < 0\}, \quad B \subseteq \{\ell_B \geq 0\}.$$



finitely many arrangement cells and outside witnesses

Figure 5: Fixed-arrangement separation. Every old atom is a finite union of arrangement cells. Outside cells and one outside witness from each split atom are separated from the convex selector by finitely many weak halfspaces.

Define

$$P_\alpha = Q \cap \bigcap_{B \in \mathcal{F}} \{\ell_B \leq 0\}.$$

This is a bounded polytope with nonempty interior, because it contains the nonempty open set U . Moreover,

$$\text{int } P_\alpha = \text{int } Q \cap \bigcap_{B \in \mathcal{F}} \{\ell_B < 0\},$$

so $U \subseteq \text{int } P_\alpha$.

If $\tau \in \mathcal{D}_+$, the whole atom lies in U and hence in $\text{int } P_\alpha$. If $\tau \in \mathcal{D}_-$, every atom point lies in an excluded face cell. If $\tau \in \mathcal{D}_\pm$, the point p_τ lies inside the new polytope while q_τ is excluded. Thus the inside/outside behavior on every old atom is exactly the same as for U . $\square$

Remark 6.3 (Tangent and lower-dimensional cells). The proof never requires a positive distance between an outside cell and U . A tangent cell is separated weakly, with strict inequality only on the open set U . Relatively open lower-dimensional cells are treated as convex sets in the ambient separation theorem. Thus degeneracy is not hidden inside an approximation argument.

Corollary 6.4 (Selector-transfer reformulation). *For any prescribed one-neuron cover of a lower code, local cover lifting is equivalent to the existence of some polytopal realization of the lower code over which an open convex selector has the prescribed atomwise behavior. Once such a selector exists, Theorem 6.2 makes it polytopal.*

Remark 6.5 (What the theorem does not say). The theorem does not transfer a selector from a curved realization of the lower code to an unrelated polytopal realization. Any obstruction must therefore be a global realization-space or repeated-neuron compatibility obstruction, not essential curvature of one neuron over a fixed finite arrangement.

7 Finite protected inner-polytopal supercodes

The previous sections reduce the universal problem to deletions. We now show that every arbitrary source realization has a finite polytopal approximation on the correct side of the problem: it preserves every target word exactly and creates only smaller words beneath source words.

Theorem 7.1 (Protected inner-polytopal supercode). *Let*

$$\mathcal{C} = \text{code}(U_1, \dots, U_n; X)$$

be a finite open convex code in a finite-dimensional affine space, and choose one atom witness x_ρ for every $\rho \in \mathcal{C}$. After deleting neurons that occur in no codeword, there exist a compact convex polytope $Q \Subset X$ and nonempty compact convex polytopes

$$P_i \Subset U_i \cap \text{int } Q$$

such that, for

$$\mathcal{D} = \text{code}(\text{int } P_1, \dots, \text{int } P_n; \text{int } Q),$$

one has

$$\boxed{\mathcal{C} \subseteq \mathcal{D} \subseteq \Delta(\mathcal{C}).}$$

Every x_ρ remains an exact witness for ρ , and $\mathcal{C}$ and $\mathcal{D}$ have the same maximal words. Deleted trivial inactive neurons may be restored by choosing lower-dimensional polytopes with empty ambient interior.

Proof. First delete every neuron that belongs to no codeword. This does not alter any nontrivial trunk, and such neurons may be restored at the end by polytopes with empty interior in the common ambient. For each $\rho \in \mathcal{C}$, openness of

$$X \cap \bigcap_{i \in \rho} U_i$$

provides a full-dimensional compact simplex S_ρ with

$$x_\rho \in \text{int } S_\rho, \quad S_\rho \Subset X \cap \bigcap_{i \in \rho} U_i.$$

The simplex is not required to lie in the atom A_ρ ; excluded neurons may cover part of it. Define

$$P_i = \text{conv} \bigcup_{\rho \ni i} S_\rho.$$

Every simplex in this union lies in the open convex set U_i , so $P_i \subset U_i$. Compactness gives $P_i \Subset U_i$. The union of the finitely many P_i and protected witnesses is compact in the open convex set X . Hence one may choose a compact polytope $Q \Subset X$ whose interior contains every P_i and every witness.

If $i \in \rho$, then

$$x_\rho \in \text{int } S_\rho \subseteq \text{int } P_i.$$

If $j \notin \rho$, then $x_\rho \notin U_j$, while $P_j \subset U_j$, so $x_\rho \notin P_j$. Thus every target word survives exactly.

For arbitrary $x \in \text{int } Q$, let

$$d(x) = \{i : x \in \text{int } P_i\}, \quad c(x) = \{i : x \in U_i\}.$$

Since $\text{int } P_i \subset U_i$, one has $d(x) \subseteq c(x)$. The source word $c(x)$ belongs to $\mathcal{C}$, so $d(x) \in \Delta(\mathcal{C})$. This proves $\mathcal{D} \subseteq \Delta(\mathcal{C})$. Every maximal word of $\mathcal{C}$ survives, while every word of $\mathcal{D}$ is contained in a source word; hence the maximal words agree. $\square$

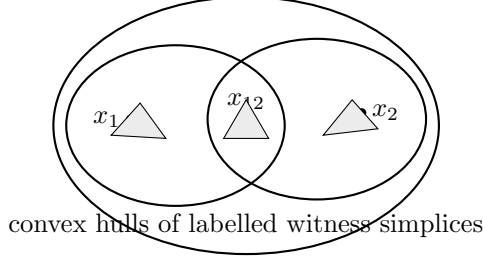


Figure 6: Protected supercode construction. Each target word has a small source-contained simplex. Neuron i is the convex hull of all simplices whose labels contain i . The selected atom witnesses remain exact.

Remark 7.2 (Degenerate atoms). A source atom may be lower dimensional or supported on several boundaries. The construction requires only that its witness lies in the open intersection of the neurons that are active there. The small simplex may cross excluded-neuron boundaries; inactive membership at the protected witness is nevertheless preserved because every new neuron polytope remains inside its source neuron.

Remark 7.3 (Finite but not exact). The theorem replaces arbitrary curved fields by finitely many polytopes but generally produces a supercode. The only errors are words obtained by deleting memberships from source words. This one-sidedness is what makes the repair atlas in the next section possible.

8 Compact whole-word repair atlases

Let $\mathcal{D}$ be the protected polytopal supercode from Theorem 7.1. For a nonempty word σ , define its complete closed carrier

$$K_\sigma = Q \cap \bigcap_{i \in \sigma} P_i.$$

Using the closed carrier rather than only the open atom ensures that threshold and lower-dimensional copies are included.

Theorem 8.1 (Compact whole-word repair atlas). *Let $\sigma \notin \mathcal{C}$ be nonempty. If $K_\sigma \neq \emptyset$, then there is a finite family*

$$\{(R_r, \rho_r)\}_{r=1}^{N_\sigma}$$

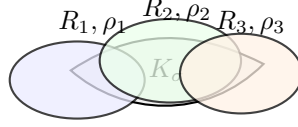
such that:

- (i) $\rho_r \in \mathcal{C}$ and $\sigma \subsetneq \rho_r$;
- (ii) R_r is a full-dimensional compact polytope with

$$R_r \subseteq \text{int } Q \cap \bigcap_{i \in \rho_r} U_i;$$

- (iii) the relative interiors cover the complete carrier,

$$K_\sigma \subseteq \bigcup_r \text{int } R_r;$$



finite source-contained whole-word cover

Figure 7: A complete unwanted carrier is compact. Source words strictly above it provide an open cover by whole-word neighborhoods; compactness yields a finite repair atlas.

(iv) *every point of K_σ is assigned at least one complete source word strictly above σ .*

The finitely many words in $\mathcal{D} \setminus \mathcal{C}$ admit one finite combined atlas.

Proof. Because $P_i \subseteq U_i$ for all $i \in \sigma$, the compact set K_σ lies in

$$\text{int } Q \cap \bigcap_{i \in \sigma} U_i.$$

For $x \in K_\sigma$, let $\rho(x)$ be its source word. Then $\sigma \subseteq \rho(x)$. Since σ is absent from $\mathcal{C}$, the containment is strict. The open set

$$\text{int } Q \cap \bigcap_{i \in \rho(x)} U_i$$

contains x , so it contains a full-dimensional compact polytope R_x with $x \in \text{int } R_x$. The interiors of these R_x form an open cover of the compact carrier K_σ . A finite subcover yields the atlas.

There are finitely many unwanted words, so the union of their finite atlases is finite. $\square$

Corollary 8.2 (One-label local deletion). *For each patch choose any index $j_r \in \rho_r \setminus \sigma$ and define*

$$\hat{P}_j = \text{conv} \left(P_j \cup \bigcup_{r: j_r = j} R_r \right).$$

Leaving the P_i with $i \in \sigma$ unchanged eliminates the word σ , keeps every enlarged polytope inside its source neuron, and preserves all protected target witnesses.

Proof. Every set used to enlarge P_j is compactly contained in the open convex set U_j , so its finite convex hull remains in U_j . A point in all P_i for $i \in \sigma$ belongs to K_σ and hence to the interior of some R_r , which activates an index $j_r \notin \sigma$. Protected positive memberships cannot be lost; a protected inactive witness cannot enter $\hat{P}_j \subset U_j$. $\square$

Remark 8.3 (Elimination of local analytic infinity). No positive separation or nondegeneracy is assumed. Even if a forbidden carrier accumulates on many source boundaries, compactness produces finitely many full-dimensional source-contained charts. Thus zero margin does not force an infinite local atlas.

Remark 8.4 (Why the theorem stops short). The corollary eliminates one current carrier by assigning patches to selected neurons. Applying many such operations, or convexifying all occurrences of a repeated neuron independently, may create new mixed-hull intersections elsewhere. The existence of a finite atlas is not the existence of one simultaneous global convex attachment.

9 Why naïve globalization fails

The preceding results reduce arbitrary curved data to finitely many polytopes and finitely many whole-word repair patches. The final step is not automatic. This section gives an exact rational calibration showing why repairs must be synchronized by complete words.

Example 9.1 (Sequential repair can recreate a forbidden word). Work in the open square $X = (-3, 3)^2$. The source fields are

$$U_1 = U_3 = X,$$

and U_2 is the interior of

$$P_2 = \text{conv}\{(0, -1), (1, 1/10), (2, 1/10), (2, -1/10)\}.$$

The covered source code is

$$\mathcal{C} = \{13, 123\}.$$

Let

$$P_3^0 = [-2, 2] \times [-1/10, 1/10]$$

and

$$P_1^0 = \text{conv}\{(-2, -1/10), (2, -1/10), (2, 1/10), (0, 1), (-2, 1/10)\}.$$

The points

$$(-3/2, 0) \in A_{13}, \quad (3/2, 0) \in A_{123}$$

are protected target witnesses. The tuple also realizes forbidden words: $(0, 4/5)$ has word 1, and $(1/5, -4/5)$ has word 2.

First replace P_3^0 by P_1^0 . Then every membership in neuron 1 is accompanied by neuron 3, so words 1 and 12 disappear.

Next attempt to repair word 2 using neuron 1 alone. Replace P_1^0 by

$$P^{\text{hull}} = \text{conv}(P_1^0 \cup P_2).$$

The exact point

$$q = (-1, -3/10)$$

lies in $\text{int } P^{\text{hull}}$ but outside both P_2 and the unchanged $P_3 = P_1^0$. Hence the forbidden word 1 reappears.

In contrast, the synchronized replacement

$$P_1 = P_3 = P^{\text{hull}}$$

realizes exactly $\{13, 123\}$ on $\text{int } P^{\text{hull}}$: points outside P_2 have word 13, and points inside P_2 have word 123.

Proposition 9.2 (The finite compatibility problem is genuine). *The atlas of Theorem 8.1 cannot in general be globalized by a fixed ordering of independent per-neuron convex-hull enlargements. Even when every patch is source-contained and every target witness is protected, a later enlargement may create a forbidden atom outside the patch that motivated it.*

Proof. The first repair in Theorem 9.1 removes the forbidden word 1. The second enlargement is source-contained, repairs a different current defect, and nevertheless creates a new 1-atom at the rational point q . Thus neither source containment nor monotone enlargement gives global monotonicity of the atom code. $\square$

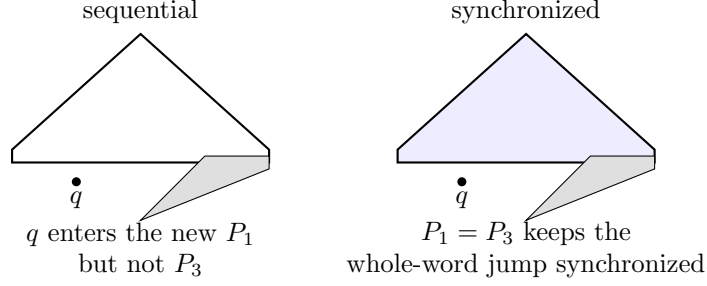


Figure 8: Sequential versus synchronized repair in Theorem 9.1. Enlarging neuron 1 alone creates a remote 1-atom; enlarging neurons 1 and 3 by the same whole-word hull preserves their synchronized activity.

9.1 A sharper one-round failure

For completeness, the reproducibility package also includes a three-neuron example with source code

$$\{\emptyset, 1, 2, 123\}$$

in which a single 123-patch covers the entire old 12-carrier, is added simultaneously to all three neurons, and yet a remote rational point still has word 12. A second larger common-core patch removes the circuit. This example shows that “add every patch to every neuron in its label in one round” is also not a theorem. We keep it in the appendix because the simpler sequential example already isolates the conceptual issue.

9.2 Other shortcuts ruled out

Three additional cautions delimit the scope of our results.

- (i) **Nerve preservation is insufficient.** A nerve-preserving approximation may expose a smaller atom, even when every intersection remains nonempty.
- (ii) **Witness preservation is insufficient.** Keeping one point from every target atom proves only $\mathcal{C} \subseteq \mathcal{D}$; it does not rule out $\mathcal{D} \setminus \mathcal{C}$.
- (iii) **Intersection completion is not a retract shortcut.** A surjective image of an intersection-complete code is intersection complete [5, Theorem 1.5]. Therefore a non-intersection-complete code cannot be obtained as a morphic retract of its intersection completion; see Section C.

The remaining task is neither uncontrolled approximation nor an infinite local repair problem. It is a finite global compatibility problem: attach a finite collection of whole-word modules while each original neuron is represented by one convex polytope and no mixed convex combination produces a forbidden intermediate word.

10 Discussion

The theorem suite gives two complementary normal forms for the convex-to-polytope conjecture.

The first is *local and rational*. Universal polyhedralization reduces to binary-meet deletion. Every such deletion has a finite rational interval bridge contained in the strict trunk above the

missing word. If an appropriate selector exists over a fixed polytopal lower realization, finite cell separation makes the selector polytopal in the same dimension. Hence neither a continuum of bridge events nor curvature of one selector over a fixed arrangement is the obstruction.

The second is *global and finite*. Every source realization has a protected inner-polytopal supercode with exactly the same maximal words and a finite set of unwanted atoms. Every complete unwanted carrier has a finite cover by source-contained polytopes labelled by complete target words. Thus degeneracy and zero margin do not force an infinite collection of local charts. The obstruction is the simultaneous convex globalization of repeated neuron labels across those finitely many modules.

10.1 Oriented-matroid interpretation

If the remaining globalization theorem is proved, the final receptive fields have finitely many facets. Take one affine hyperplane for every facet of every neuron polytope and of the ambient polytope. The resulting finite real hyperplane arrangement represents an oriented matroid M . Each original neuron is the intersection of the positive halfspaces corresponding to its facets, hence is represented by a trunk of $L^+(M)$. Restricting to the ambient trunk and reading those neuron trunks gives a surjective code morphism

$$L^+(M) \twoheadrightarrow \mathcal{C}.$$

This is the constructive content behind the characterization of Kunin et al. [8].

Conversely, a negative answer would require more than the failure of a chosen approximation, cap, bridge, or finite lift. One would need an explicit convex code whose every finite representable completion fails. None of the exact failure examples in this paper has that invariance: each obstructs a particular globalization rule, not polytope convexity of the code.

10.2 The remaining theorem

The unresolved statement can be formulated without any limiting language.

Finite simultaneous attachment problem. Given a protected inner-polytopal supercode $\mathcal{D}$ of an open convex code $\mathcal{C}$ and one finite source-contained whole-word repair atlas for every word in $\mathcal{D} \setminus \mathcal{C}$, construct finitely many polytopes, possibly in a larger dimension, that activate each atlas label synchronously, eliminate every unwanted carrier, preserve every protected target word, and represent every repeated original neuron by one convex polytope.

A solution would imply universal polyhedralization directly. By Theorem 4.5, it would also solve the canonical binary-meet deletion operation. The rational bridge theorem supplies the one-dimensional interface data required in that local formulation.

10.3 Algorithmic scope

All objects constructed in the present reductions are finite once an input realization and atom witnesses are given. This does not by itself yield a decision algorithm for convexity: the input realization is not combinatorial data, and no computable universal bound on a final completion is proved. Likewise, a bounded search for oriented-matroid completions cannot certify nonpolytopality without a separate completion-size bound.

10.4 Biological interpretation

The code records exact co-firing regions rather than only intersections of receptive fields. Replacing smooth place fields by polytopes would show that the finite combinatorics of co-firing never requires curved receptive-field boundaries. The present results establish this in the plane and for the finite classes above, and show that the unresolved issue is how several appearances of the same receptive field couple across a finite collection of local repair modules. We make no claim that biological receptive fields are polytopal, only that their finite atom codes may be.

In summary, the conjecture has been reduced to a canonical binary-meet deletion with finite rational bridge data and, independently, to a finite simultaneous globalization problem for protected whole-word repair atlases. These reductions eliminate uncontrolled local curvature and infinite local repair as possible explanations for the difficulty, but they do not settle the conjecture.

A Convention crosswalk

The literature uses closely related minor conventions. We record only the translations used in this paper.

Lemma A.1 (Operation-by-operation crosswalk). *For surjective morphisms between codes already containing $\emptyset$, the older and mandatory-empty-word definitions agree. An older morphic image with constant-zero or constant-one output coordinates may first delete those coordinates; after adjoining $\emptyset$, the morphism extends by $f(\emptyset) = \emptyset$. Replacing a code $\mathcal{A}$ by a trunk $T = \text{Tk}_{\mathcal{A}}(\alpha)$ corresponds to the mandatory-empty-word morphism*

$$r_T : \mathcal{A} \cup \{\emptyset\} \longrightarrow T \cup \{\emptyset\}, \quad r_T(c) = \begin{cases} c, & c \in T, \\ \emptyset, & c \notin T. \end{cases}$$

Thus the representable-oriented-matroid characterization may be used under the convention of this paper after removal of constant coordinates.

Proof. For a morphism between mandatory-empty-word codes, surjectivity to $\emptyset$ rules out full inverse images of proper target trunks, and nonemptiness of a target trunk rules out empty inverse images. Conversely, the empty and full trunks have empty and full inverse images automatically. Constant coordinates are isomorphism-level redundancies in the older convention. For trunk replacement, the inverse image of a proper trunk rooted at β is the trunk rooted at $\alpha \cup \beta$; the map is surjective and fixes the words of T . $\square$

B Cap-construction audit

We collect the edge cases in Theorem 3.1.

Several apices. A point of Q' may use many apices. The implication $z_s \geq h_s \Rightarrow b_s \geq \lambda_s > 1/2$ is derived separately for each cap. It follows that two cap inequalities cannot hold simultaneously, independent of all smaller apex coefficients.

Threshold equality. At $z_s = h_s$, the preliminary projection is still in the open maximal atom. Cut neurons are absent because the point lies on the boundary of their final polytope; uncut active neurons remain interior. Hence the equality layer carries σ_s rather than a third word.

Two requested words under one maximal word. Their caps use independent height coordinates, even when $m_s = m_t$. Pairwise disjointness follows from the apex coefficients, not from disjointness of the source balls.

Nested requested words. No inclusion assumption among the σ_s is used. A point lies in at most one cap, so nested cuts do not interact.

Lower-dimensional old atoms. Only maximal atoms are used for cap projection. These are open in the ambient. Existing lower-dimensional atoms are preserved at height zero and are not used to obtain source balls.

Empty and full neurons. An always-inactive neuron is removed and restored by a lower-dimensional polytope. An always-active neuron is represented by the ambient polytope and is cut in a cap exactly when it is absent from the requested word.

Empty atom. The chosen empty-word witness remains at height zero. No cap reaches it because every cap projects to a nonempty maximal atom.

Finiteness. The construction introduces exactly one coordinate and at most one new halfspace in each affected neuron for every added word. All polytopes have finitely many facets.

C Intersection completeness under morphisms

The following fact is known from Jeffs [5, Theorem 1.5]. We include a convention-compatible proof because it closes a tempting shortcut.

Proposition C.1 (Intersection-morphism theorem). *If $\mathcal{D}$ is intersection complete and*

$$f : \mathcal{D} \twoheadrightarrow \mathcal{C}$$

is a surjective code morphism, then for every finite nonempty family $d_1, \dots, d_m \in \mathcal{D}$,

$$f \left(\bigcap_{a=1}^m d_a \right) = \bigcap_{a=1}^m f(d_a).$$

Consequently $\mathcal{C}$ is intersection complete.

Proof. For each output neuron j , let

$$T_j = f^{-1}(\text{Tk}_{\mathcal{C}}(j)).$$

If $T_j = \emptyset$, coordinate j is constantly absent. Otherwise, intersection completeness gives a principal root

$$r_j = \bigcap_{d \in T_j} d$$

with $T_j = \text{Tk}_{\mathcal{D}}(r_j)$; a full trunk is represented by $r_j = \emptyset$. Coordinatewise,

$$\begin{aligned} j \in f\left(\bigcap_a d_a\right) &\iff r_j \subseteq \bigcap_a d_a \\ &\iff r_j \subseteq d_a \text{ for every } a \\ &\iff j \in f(d_a) \text{ for every } a. \end{aligned}$$

Surjectivity then shows that finite intersections of target words lie in $\mathcal{C}$. □

Corollary C.2 (No intersection-completion retraction). *A non-intersection-complete code is not a surjective morphic image of an intersection-complete code. In particular, it is not a morphic retract of its intersection completion.*

This corollary explains why the polytopality of $\widehat{\mathcal{C}}$ cannot be followed by a purely categorical retraction to an arbitrary $\mathcal{C}$.

D Exact finite examples and verifier scope

The accompanying scripts use exact rational arithmetic. They check finite calibrations; they do not replace the universal proofs.

D.1 Rational interval events

The verifier represents interval endpoints by rational numbers and evaluates membership on every endpoint class and complementary open cell. It includes simultaneous starts and ends, empty intervals, full intervals, and marked endpoint words.

D.2 Fixed-arrangement cells

Small rational arrangements include a tangent cell, a lower-dimensional cell, a wholly outside atom, and a split atom with an outside witness on a separator boundary. Exact halfspace signs confirm the theorem’s classifications.

D.3 Sequential and synchronized repair

For Theorem 9.1, exact polygon membership confirms:

1. the protected words 13 and 123 occur;
2. the first synchronized equality $P_1 = P_3$ removes word 1;
3. $(-1, -3/10)$ has word 1 after the later one-neuron hull;
4. $P_1 = P_3 = P^{\text{hull}}$ realizes exactly $\{13, 123\}$.

D.4 One-round failure

The optional Phase-XIX calibration verifies a three-neuron source code $\{\emptyset, 1, 2, 123\}$, a protected supercode with an unwanted 12 atom, a complete 123 repair patch covering the old carrier, and a remote rational 12 witness after the one-round synchronized hull. A second common-core polytope eliminates the defect. This example is included to falsify an overly strong globalization conjecture, not as a component of any universal proof.

D.5 Scope statement

The general theorems are human-readable mathematical proofs. The scripts certify only the stated finite coordinate examples, combinatorial deletion orders, and rational endpoint data. Failure of a finite search or a particular ansatz is nowhere used as evidence of nonpolytopality.

E Logical scope, finiteness, and quantitative bookkeeping

This appendix records the exact quantifiers in the three universal reductions and the finiteness information supplied by the constructive theorems. It is included to prevent two common overreadings: that a reduction is itself an algorithm on abstract codes, or that a finite local atlas already provides a simultaneous global realization.

E.1 Theorem-scheme equivalences

Let **UP** denote the statement that every finite open convex code is polytope convex. Let **CL** denote the universally quantified cover-lifting statement from Theorem 4.3, **DL** the one-word deletion statement from Theorem 4.4, and **BM** the binary-meet deletion statement from Theorem 4.5. Then

$$\mathbf{UP} \iff \mathbf{CL} \iff \mathbf{DL} \iff \mathbf{BM}.$$

The reverse implications from **UP** are formal: the upper code in each local statement is assumed open convex, so **UP** makes it polytopal. The forward implications are constructive inductions on finite code data:

- (a) **CL** $\Rightarrow$ **UP** follows along a saturated chain in the finite interval below a fixed code;
- (b) **DL** $\Rightarrow$ **UP** starts from the maximal-intersection completion and deletes missing nonmaximal words one at a time;
- (c) **BM** $\Rightarrow$ **DL** orders the missing maximal-word intersections by nondecreasing cardinality and uses an inclusion-minimal family of maximal words to express each deleted word as a meet of two strict survivors.

These are equivalences of universal statements, not a claim that arbitrary immediate covers, arbitrary deletions, and binary-meet deletions are the same geometric operation. The binary-meet formulation is preferred because open convexity supplies the rational interval bridge of Theorem 5.2.

E.2 Finite construction bounds

The present proofs provide the following explicit finiteness data.

Construction	Added dimension	Finite data
Polytope monotonicity	$ \mathcal{E} \setminus \mathcal{A} $	one apex and one threshold per added word
Maximal-intersection core	at most $k - 1$ for k maximal words	one coordinate halfspace for each missing maximal-word incidence
Rational bridge	one-dimensional trace; optional simplex thickening	at most two endpoints per nontrivial relative neuron, modulo equality classes
Fixed-arrangement selector	no dimension increase	finitely many old arrangement cells and one separator per outside cell/witness
Protected inner supercode	no dimension increase	one compact simplex per source word
Whole-word repair atlas	no dimension increase locally	a finite subcover for each compact unwanted carrier

No uniform numerical bound on the last atlas size is asserted from the abstract code alone, because its extraction uses a supplied convex realization. Likewise, none of these bounds gives a computable bound on the size of a representable-oriented-matroid completion for every open convex code.

E.3 Two finite objects that must not be conflated

The rational bridge and the whole-word atlas solve different local problems.

The bridge is attached to one missing binary meet $\sigma = \tau \cap \nu$. It records a finite covered path of strict superwords between two protected endpoint words. It supplies interface data but does not select a compatible placement in a pre-existing polytopal arrangement.

The atlas is extracted after choosing a protected inner-polytopal supercode. It covers the complete compact carrier of every unwanted word by source-contained modules labelled by complete target words. It removes the possibility that zero margin requires infinitely many local charts, but modules carrying the same neuron may occur in several geometrically separated places. Convexifying those occurrences independently can create remote mixed words, as shown in Theorem 9.1.

Thus the final open statement is finite in two independent senses: there are finitely many unwanted carriers, and each has finitely many complete-word modules. The missing theorem is compatibility of their simultaneous convex attachment, not extraction of further local data.

F Verification specifications and proof–computation boundary

The reproducibility package accompanying this paper has two distinct roles. It is not a formal proof assistant for the universal statements. Instead, it provides deterministic exact-arithmetic replay for the finite coordinate examples and for combinatorial identities that are especially vulnerable to strict-versus-weak inequality mistakes. The general theorems are certified by the mathematical arguments in the main text.

F.1 Exact interval traces

A rational interval trace is encoded by a finite family of open intervals

$$J_j = (a_j, b_j) \subset T = (A, B)$$

with rational endpoints. The verifier forms the finite ordered partition of T determined by all endpoint equality classes. It evaluates the relative word on every complementary open interval and at every endpoint class. This catches the two errors that are invisible to generic sampling: a simultaneous start/end event that is accidentally ordered, and an endpoint word that changes when strict inequalities are replaced by weak ones. Rationalization is checked by comparing the complete ordered-partition word table before and after replacing the source endpoint values.

F.2 Finite arrangement separation

The fixed-arrangement calibration uses rational halfspaces and checks four cell types: a relatively open full-dimensional cell wholly outside the selector, a split cell, a lower-dimensional cell, and a tangent cell whose closure meets the selector. For each retained separator, the script verifies that every protected inside point satisfies the inequality strictly and that the designated outside witness violates it weakly. The finite intersection is then evaluated on one representative from every arrangement cell and every lower-dimensional face used in the example.

This computation illustrates, but does not replace, the proof of Theorem 6.2. In the theorem, finiteness comes from the old facet arrangement, and separation is invoked once for each wholly outside cell and once for a chosen outside witness in each split cell.

F.3 Repair-atlas calibrations

The planar sequential-repair example is checked using exact rational polygon membership. The script verifies the protected words 13 and 123, the disappearance of word 1 after the first synchronized equality $P_1 = P_3$, the reappearance of word 1 at

$$(-1, -3/10)$$

after a later one-neuron hull, and the exact final code $\{13, 123\}$ under the synchronized replacement $P_1 = P_3 = P^{\text{hull}}$.

The optional one-round calibration uses exact rational convex hulls and branchwise strict linear feasibility. It verifies a source code $\{\emptyset, 1, 2, 123\}$, an inner-polytopal supercode with one unwanted 12 carrier, a complete 123 atlas patch covering the old carrier, and the remote forbidden witness $(0, 1)$ after one-round synchronized hulls. A second common-core polytope is then checked to restore the exact target code. This example is included only to demonstrate that finite simultaneous attachment is nontrivial; no finite-stage failure is interpreted as a nonpolytopality obstruction.

F.4 Manifest and clean-build policy

The release contains a SHA-256 manifest over all source, verifier, metadata, and outreach files. The replay script first checks this manifest, runs the exact examples, compiles the manuscript and this appendix from source, and compares the resulting artifact hashes against the release ledger. The arXiv tarball is separately extracted and compiled in a clean directory with no network access and no private path dependencies.

The proof-computation division is therefore explicit:

Claim type	Certification method
Universal geometric theorem	Human-readable proof in the manuscript
Finite rational coordinate example	Deterministic exact arithmetic
Strict/weak endpoint table	Exhaustion of endpoint classes and open cells
Fixed finite LP branch	Rational feasibility witness or Motzkin/Farkas certificate
Absence of bounded-search output	No mathematical conclusion drawn

This separation prevents the verifier from being presented as evidence for claims outside its finite scope.

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